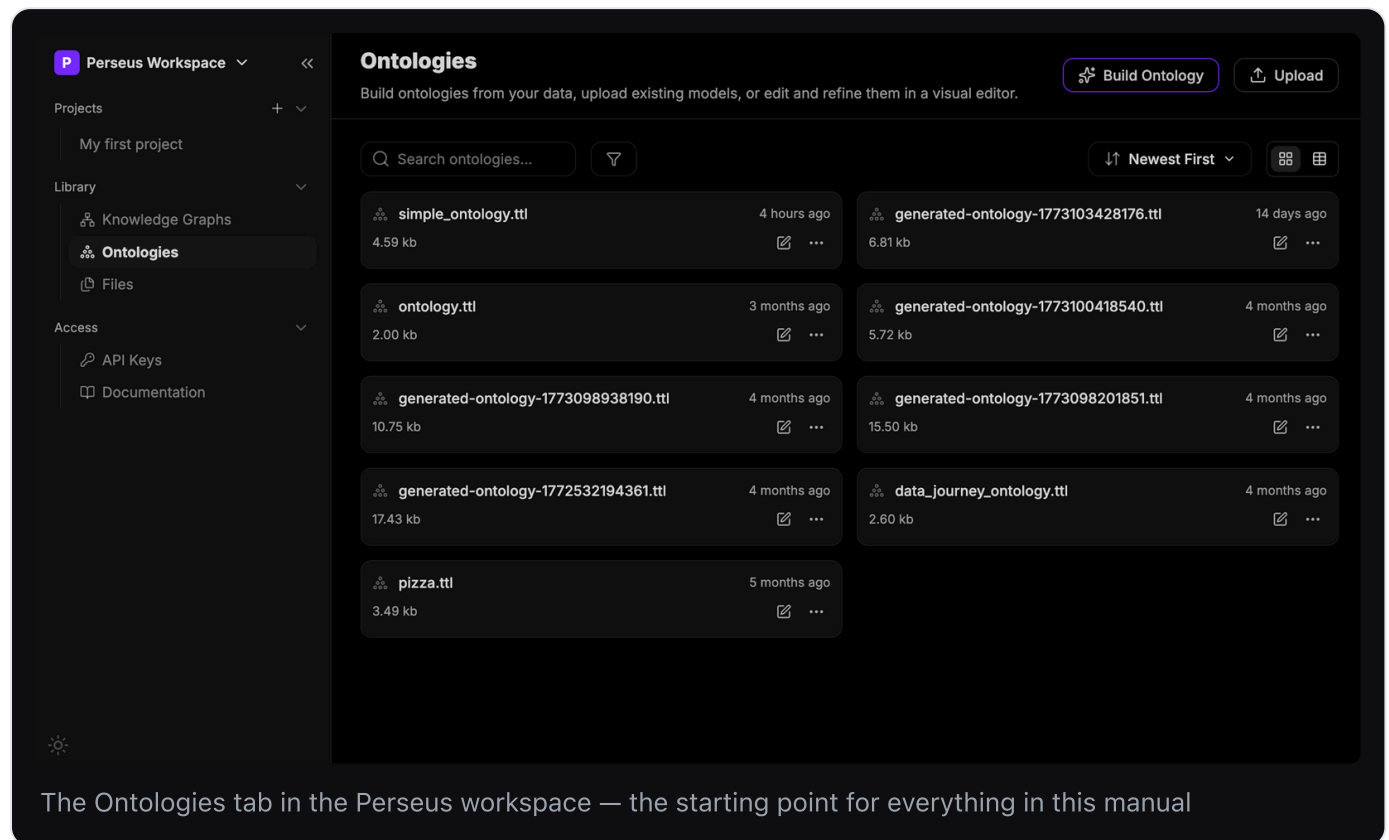


Lettria Perseus — Ontology Editor Manual

A guide to the visual ontology editor in Lettria Perseus.



1. What this tool is for

Perseus extracts structured data out of unstructured documents. Left alone, it will extract whatever it finds. An **ontology** is how you tell it what it is *allowed* to find: which kinds of things exist in your domain, what attributes they have, and how they can be connected to one another.

The ontology editor is where you inspect and hand-tune that schema. Use it to:

- Review an ontology that Perseus generated from your documents, and fix what it got wrong.
- Add a class or a relationship the generator missed.
- Rename things so the extracted graph uses your team's vocabulary.
- Seed a few known entities (individuals) so extraction has anchors to attach to.

An ontology editor is a schema editor. It is closer in spirit to editing a database schema than to editing data — except the "schema" here is itself expressed as a graph.

2. Concepts and Vocabulary

Graph

A **graph** is nodes connected by edges. In a knowledge graph the nodes are *things* (San Francisco, Jane Doe, the USA) and the edges are *labelled relationships* between them (Jane Doe `—LOCATED_IN→` San Francisco).

The unit of information is a **triple**: subject, predicate, object.

Jane Doe	LOCATED_IN	San Francisco
↑	↑	↑
subject	predicate	object

Everything in the editor — every class, every property, every annotation — compiles down to a pile of triples. The UI is a friendlier face over that.

Ontology

An **ontology** is the schema of the graph: the vocabulary and the rules. It declares that `Person` and `City` are kinds of things, that `LOCATED_IN` is a legal edge, and that it may run from a `Person` to a `City` (but not, say, from a `Year` to a `Year`).

In relational terms, an ontology is roughly your `CREATE TABLE` statements plus your foreign key constraints, except it is more expressive and it is stored as data rather than as DDL.

An ontology is *not* your data. It describes the shape of your data.

Class

A **class** is a type — a category of thing. `Person` , `City` , `Country` , `Year` . Class names are conventionally singular and PascalCase.

Classes form a hierarchy. `City` can be a child of `Place` , which can be a child of `Thing` . A child class is a **subclass**: every `City` is automatically also a `Place` . This is the "is-a" relationship from OO inheritance, and as in some OO languages, a class can have more than one parent.

Perseus displays this hierarchy in the left panel of the **Classes** tab, with an expand chevron on any class that has children.

Individual

An **individual** is one specific, named instance of a class. `San Francisco` is an individual of class `City`. `Jane Doe` is an individual of class `Person`.

Class is to individual as `class Person` is to `new Person("Jane")` — the type versus one concrete thing of that type.

You do not have to declare every individual in your ontology; extraction will discover most of them from your documents. You declare the ones you want to guarantee exist, or the ones you want to attach known facts to.

Property

A **property** is a named edge — the predicate in a triple. Perseus splits them into three kinds, the same split OWL uses, and they behave differently:

Kind	Points from	Points to	Example
Object property	an individual	another individual	<code>LOCATED_IN</code> , <code>employs</code> , <code>manages</code>
Datatype property	an individual	a literal value (string, number, date)	<code>name</code> , <code>description</code> , <code>url</code>
Annotation property	any resource	metadata about the model itself	<code>rdfs:label</code> , <code>rdfs:comment</code>

Which kind you want is decided by the right-hand side. If the thing on the right is something you would want to describe further — it has its own attributes, its own relationships — it is an **object property**. If it is just a value, a string or a number, it is a **datatype property**.

`Jane Doe LOCATED_IN San Francisco` is an object property: `San Francisco` is a real entity with its own facts. `Jane Doe name "Jane Doe"` is a datatype property: the string is just a string. `employer` pointing at a `Company` is an object property; `employerName` holding a string is a datatype property. The first is usually what you want in a knowledge graph — the links are the point of the graph.

Annotation properties are the odd one out: they carry documentation, not domain facts. They are for humans and tools, and reasoners ignore them.

Domain and Range

Every object and datatype property has a **domain** and a **range**.

- **Domain** = the class(es) the *subject* is allowed to be. The left side of the arrow.
- **Range** = the class(es) the *object* is allowed to be. The right side of the arrow.

In the sample ontology, `LOCATED_IN` has domain `{City, Country}` and range `{City, Person, Country}`. Read that as: *"the thing doing the locating can be a City or a Country; the thing it is located in can be a City, a Person, or a Country."*

Listing multiple classes in a domain or range means *any of them* is acceptable — it is a union, not an intersection.

Domain and range are how you constrain extraction. A property with a narrow domain and range is a strong signal to the extractor; a property with neither will match almost anything. Start narrow, look at the extracted graph, and widen only where you see real misses.

Annotation

An **annotation** is human-readable metadata attached to a class, property, or individual. It has no effect on the logic of the graph; it exists so that you, your teammates, and the extraction model can tell what a thing means.

The most common ones:

- `rdfs:label` — the display name. Perseus sets this for you when you name a resource.
- `rdfs:comment` / `dcterms:description` / `skos:definition` — a prose description.
- `rdfs:seeAlso`, `rdfs:isDefinedBy` — pointers to related or defining resources.

Annotations are **language-tagged**. Each annotation row has a language selector on the right (`en` (English) , or the generic `lang` if unset). This is how one ontology serves a multilingual pipeline: the same class can carry an English label and a French label.

Descriptions are load-bearing. Perseus's extraction is LLM-driven, and a class's annotations are part of what the model reads when deciding whether a span of text is an instance of that class. An `rdfs:comment` on `Person` saying *"an individual human named in the source text, not a fictional or hypothetical person"* changes what comes back. Leaving it blank leaves the model guessing.

URI / IRI

Every resource in an ontology has a globally unique identifier, written as a URL, shown in grey at the top-right of every detail panel:

```
http://www.w3.org/2002/07/owl#Country
http://example.org/ontology#employs
```

It is an *identifier*, not an address — nothing is fetched from it. It exists so that two ontologies from two different teams can be merged without `Person` colliding with `Person`.

The part after the `#` is the local name, which Perseus derives from the name you type. A freshly created, still-unnamed resource gets a placeholder URI with a timestamp in it (`owl#timestamp-1783753869409`); once you type a real name, the URI settles to `owl#Country`. That is expected. Just don't save a resource while it is still called *Untitled*.

Prefixes you will see in the dropdowns

The `rdfs:` / `owl:` / `dcterms:` prefixes in the annotation dropdown are shorthand for long URIs. Knowing what family a term comes from helps you pick the right one:

Prefix	Stands for	Use it for
<code>rdf:</code>	RDF core	the raw triple model; <code>rdf:type</code>
<code>rdfs:</code>	RDF Schema	<code>rdfs:label</code> , <code>rdfs:comment</code> , subclass/subproperty
<code>owl:</code>	Web Ontology Language	classes, properties, <code>sameAs</code> , <code>differentFrom</code>
<code>skos:</code>	Simple Knowledge Organization System	<code>skos:definition</code> , thesaurus-style vocabularies
<code>dc:</code> / <code>dcterms:</code>	Dublin Core	<code>title</code> , <code>description</code> , generic document metadata

When several of them offer near-identical terms (`rdfs:comment` vs `dcterms:description` vs `skos:definition`), pick one and be consistent across the ontology. Consistency matters more than which one you picked.

Turtle (.ttl)

Ontologies in Perseus are stored as **Turtle** files — a plain-text serialisation of RDF triples, hence the `.ttl` extension on every ontology in the library. It is readable:

```
:San_Francisco    rdf:type        :City .  
:San_Francisco    rdfs:label      "San Francisco"@en .  
:San_Francisco    :LOCATED_IN    :USA .
```

You never have to write Turtle to use the editor. But everything you do in the UI is building these lines, and if you export an ontology this is what you get.

Turtle is also your tiebreaker. If you cannot tell whether a field did what you meant — whether that Relationship row landed as a schema constraint or as an actual fact — save, then open **View TTL** from the ontology card's **...** menu and read the source.

Where WebProtégé fits in

Perseus's editor is modelled on **WebProtégé**, the browser version of Protégé, the open-source ontology editor from Stanford. The mental model transfers directly: the same Classes / Properties / Individuals split, the same domain/range semantics, the same annotation-based documentation, the same commit-with-a-message revision history.

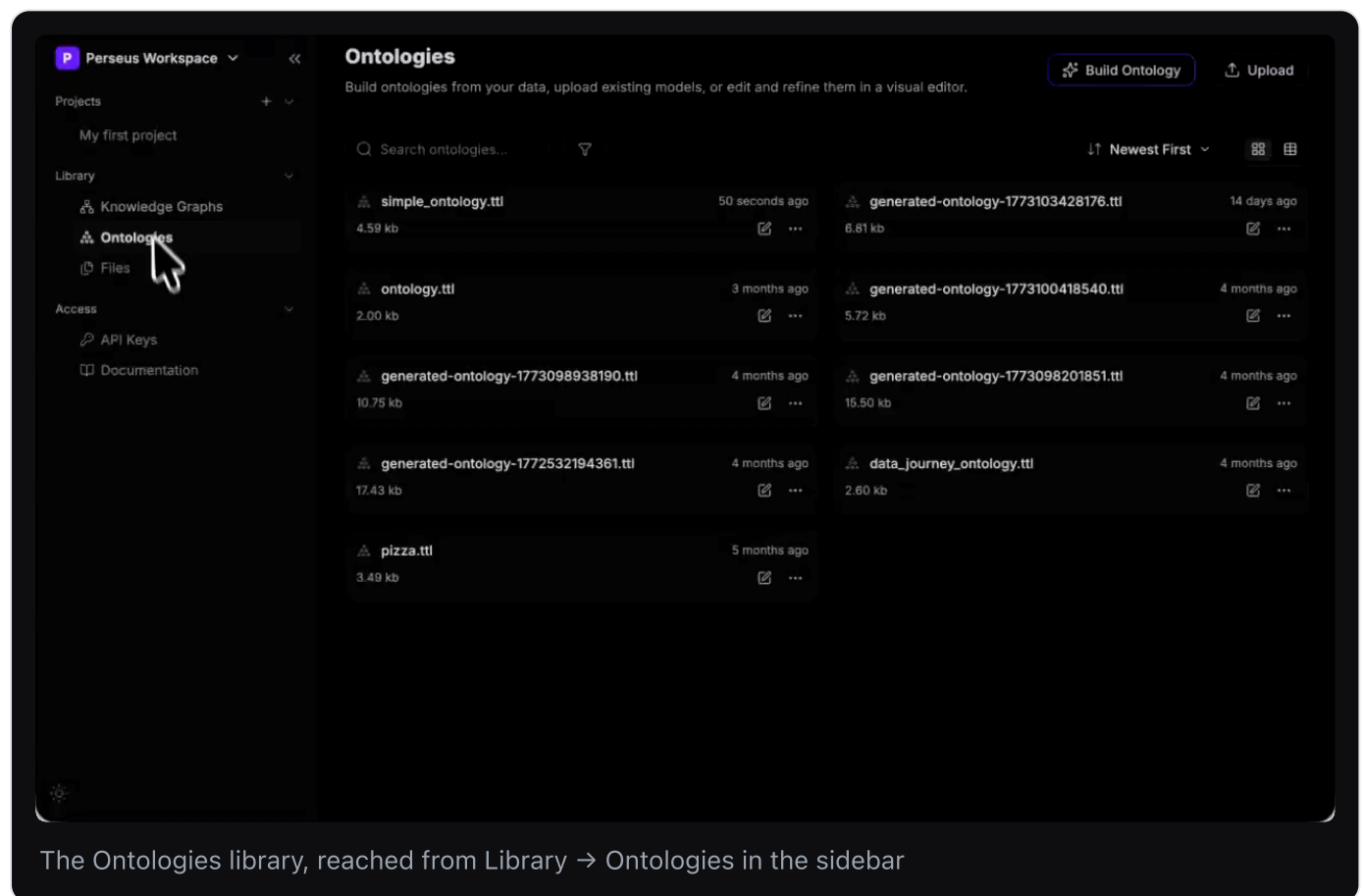
Perseus is a deliberately smaller surface. It drops WebProtégé's heavier OWL machinery (complex class expressions, restrictions, reasoner integration, discussion threads) in favour of the subset that drives Perseus extraction.

3. Getting to the editor

The Ontologies library

In the left sidebar of the Perseus workspace, under **Library**, click **Ontologies**. (The sibling entries are **Knowledge Graphs** — the extracted output — and **Files** — the source documents.)

The Ontologies page lists every ontology in the workspace as a card showing its filename, size, and last-modified time. You get a search box, a filter, a **Newest First** sort, and a grid/table view toggle.



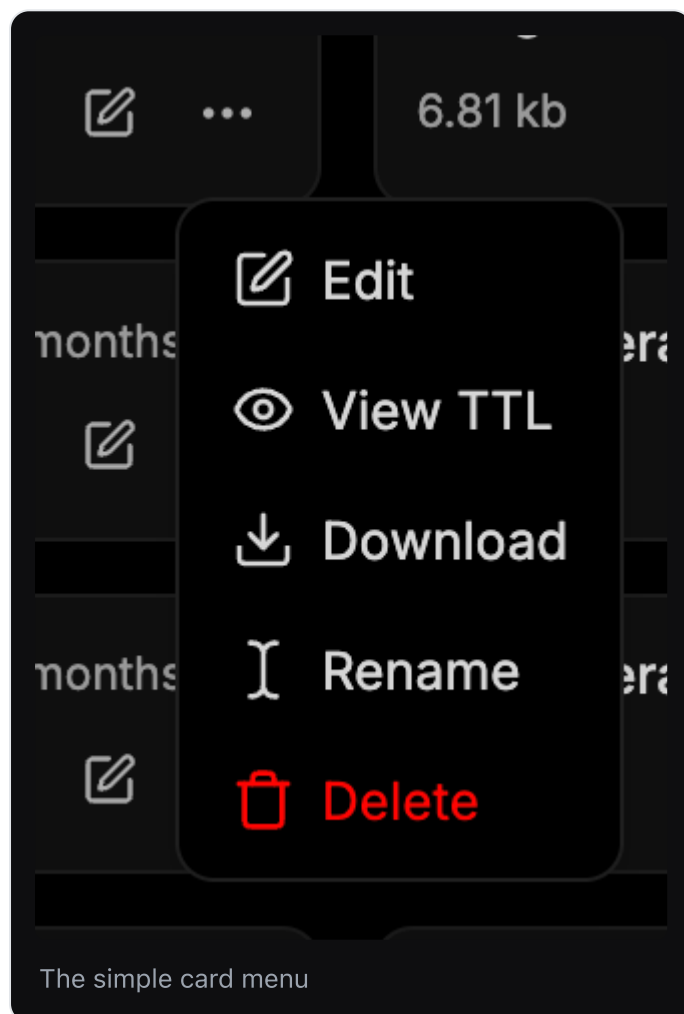
Each card has a **pencil icon** (open in the editor) and a **...** menu.

The card menu: simple and advanced

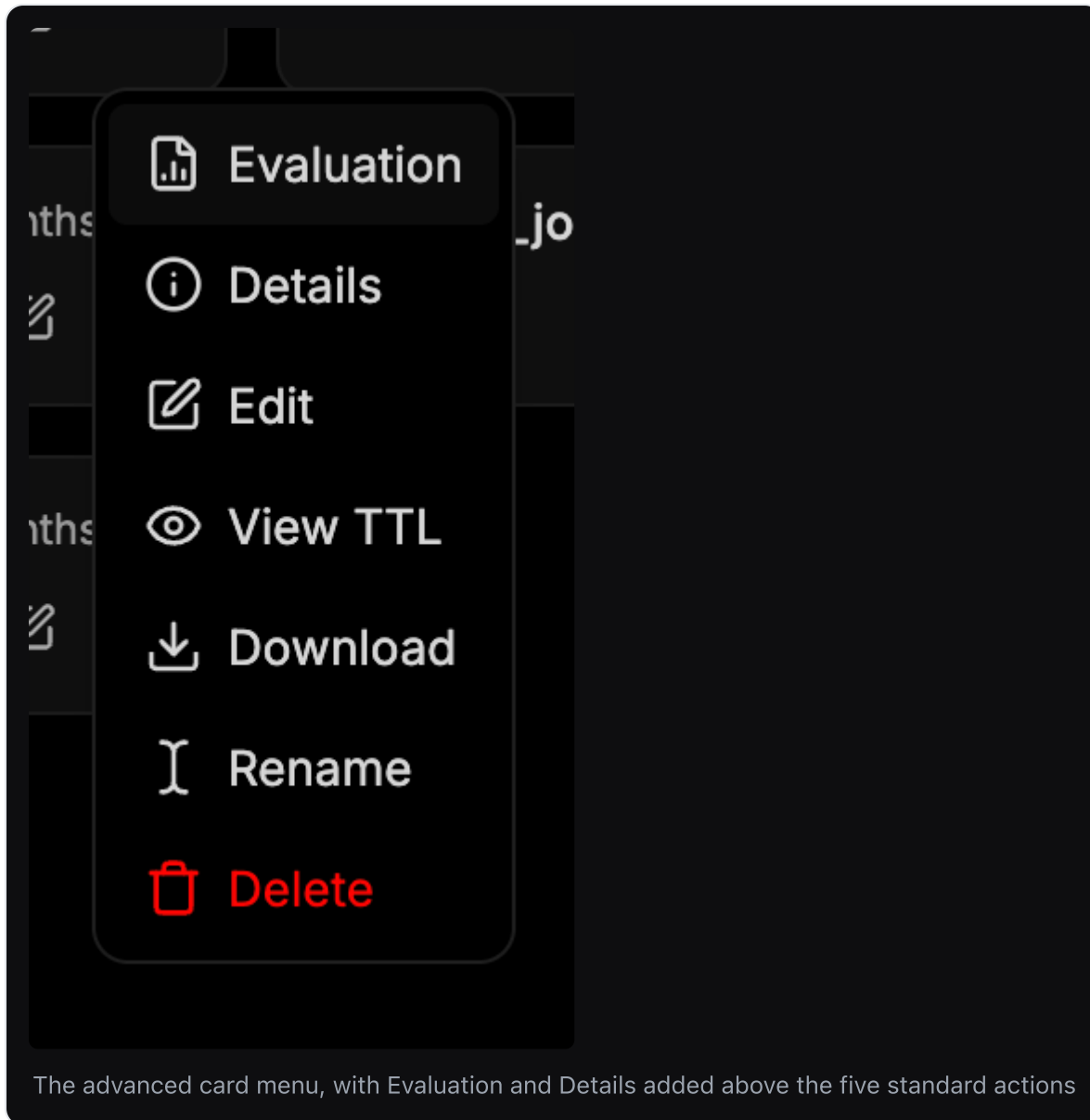
The **...** menu comes in two forms. Which one you get depends on the ontology.

The **simple menu** has five actions, and every ontology has these:

Action	What it does
Edit	Opens the ontology in the visual editor. Same as the pencil icon.
View TTL	Shows the raw Turtle source — the ground truth for what the editor has built. See Turtle (.ttl).
Download	Exports the <code>.ttl</code> file. Use this to commit an ontology to your own repo, diff two versions outside Perseus, or hand it to another tool.
Rename	Renames the file. Note this renames the <i>file</i> , not the resources inside it.
Delete	Deletes the ontology from the workspace.



The **advanced menu** adds two entries at the top — **Evaluation** and **Details**:



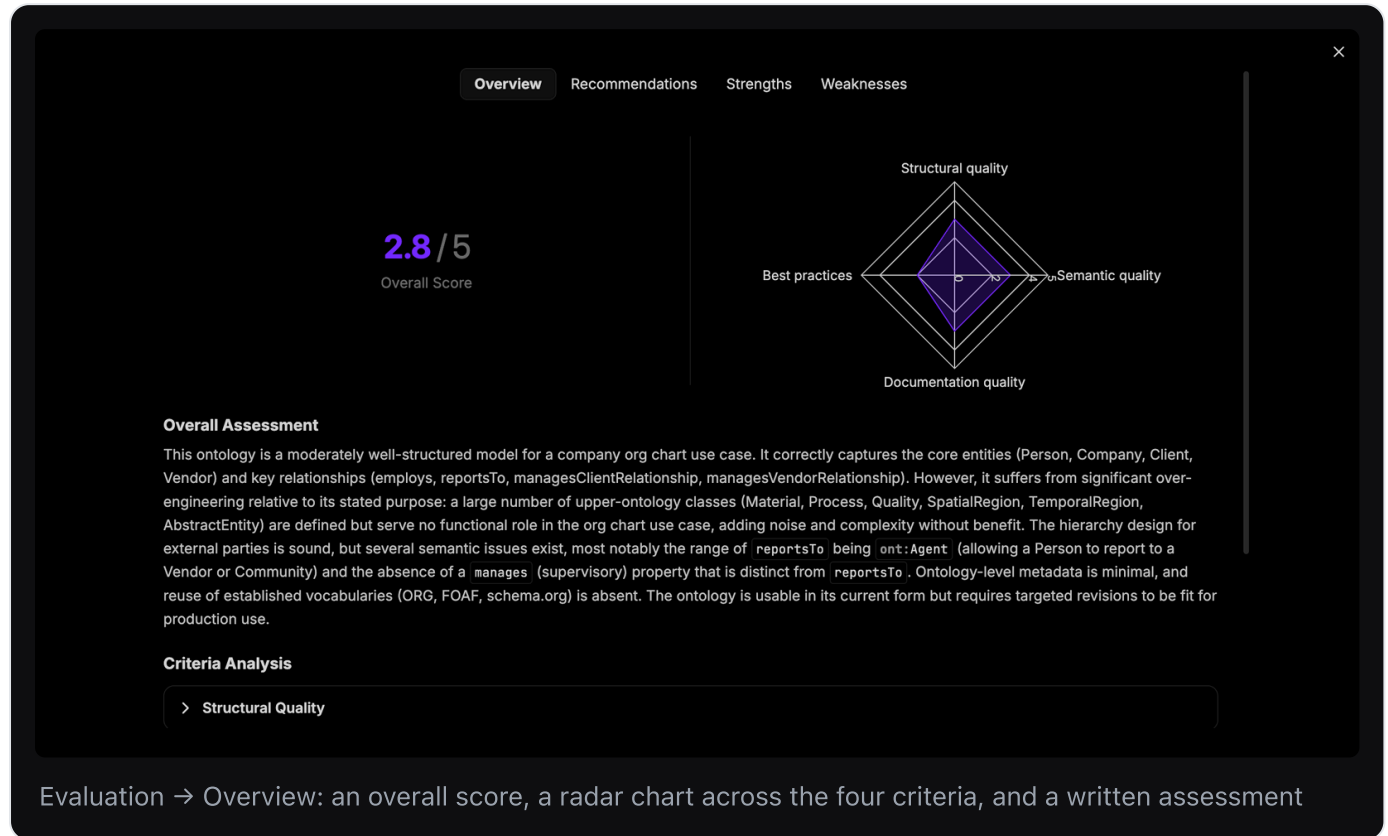
Both extras are reports about *how the ontology was produced*, not about its current contents. They appear on ontologies that Perseus generated with **Build Ontology**, because that build runs as a job and the job is what produces them: the timeline, the use case you typed, and the quality analysis. An ontology that arrived any other way — uploaded, or converted from another schema — has no such job behind it, so it gets the simple menu.

Two consequences worth knowing:

- Editing and saving an ontology does **not** refresh its Evaluation. The report describes the ontology as the build produced it, so it goes stale the moment you start editing.
- If you download a generated ontology and upload it again, the copy shows the simple menu even though the file is byte-for-byte identical. The reports belong to the build job, not to the file.

Evaluation

A quality analysis of the generated ontology, scored out of 5 across four criteria — **Structural quality**, **Semantic quality**, **Documentation quality**, and **Best practices** — shown as a radar chart with a prose overall assessment beneath it.



Four sub-tabs:

- **Overview** — the overall score, the radar chart, the written assessment, and a **Criteria Analysis** section you can expand per criterion.
- **Recommendations** — concrete, actionable changes, each with a justification. These are specific: *"Restrict the range of ont:reportsTo from ont:Agent to ont:Person", "Remove the classes ont:Material, ont:Process ... along with their disjointness axioms."* This tab is the most directly useful one — it is a work list you can take straight into the editor.
- **Strengths** — what the generator got right, and why it is right.
- **Weaknesses** — where the model is wrong or over-built, with the reasoning.

OverviewRecommendationsStrengthsWeaknesses

Add a named inverse property for `ont:reportsTo`, such as `ont:directlyManages`, with domain `ont:Person` and range `ont:Person`, and declare it as `owl:inverseOf ont:reportsTo`.

The primary use case is to 'show who works for who and who manages which clients and vendors.' Without a named supervisory property, the managerial hierarchy direction is not directly queryable, which is a fundamental gap for an org chart.

Restrict the range of `ont:reportsTo` **from** `ont:Agent` **to** `ont:Person` **(or a new class** `ont:Employee` **that is a subclass of** `ont:Person` **employed by the company).**

Allowing a Person to report to a Vendor, Community, or any Agent is semantically incorrect for an org chart. This range is too broad and would allow invalid instance data to pass without detection, undermining data integrity.

Remove or move to a separate 'upper ontology' module the classes `ont:Material`, `ont:Process`, `ont:Quality`, `ont:SpatialRegion`, `ont:TemporalRegion`, and `ont:AbstractEntity`, **along with all their associated disjointness axioms.**

These classes serve no role in the org chart use case and constitute the majority of the ontology's axiom count. Their presence obscures the core model and adds maintenance burden without any benefit to the stated purpose.

Fix the domain of `ont:hasIndustry` **from** `ont:Company` **to** `ont:Organization` **to prevent the reasoner from incorrectly inferring that any organization with an industry attribute is a Company, which would violate the Company/CommercialPartner disjointness axiom.**

As currently defined, asserting `vendorX ont:hasIndustry 'Technology'` causes a reasoner to infer `vendorX` is a `Company`, which is disjoint with `CommercialPartner` (`vendorX`'s actual class), making `vendorX`'s class membership unsatisfiable.

Evaluation → Recommendations: each recommendation names the exact resource to change and explains why

OverviewRecommendationsStrengthsWeaknesses

The Client/Vendor hierarchy under `CommercialPartner` is logically sound and correctly uses `owl:disjointWith` to prevent a single organization from being simultaneously a Client and a Vendor, which is a meaningful business constraint for the use case.

The use of `rdfs:subPropertyOf` to relate `ont:managesClientRelationship` and `ont:managesVendorRelationship` to the superproperty `ont:managesExternalRelationship` is a correct and useful design pattern, enabling general queries over all external relationship management.

Explicit inverse properties (`ont:employs` / `ont:isEmployedBy`) are correctly declared with `owl:inverseOf`, enabling bidirectional traversal of the employment relationship without redundant data entry.

The `ont:reportsTo` property is correctly annotated as `owl:AsymmetricProperty` and `owl:IrreflexiveProperty`, which are semantically appropriate constraints that prevent a person from reporting to themselves and prevent mutual reporting relationships.

All classes and properties include both `rdfs:label` and `rdfs:comment` annotations in English, providing a baseline level of human-readable documentation for every term.

Evaluation → Strengths

OverviewRecommendationsStrengthsWeaknesses

Severe over-engineering relative to the use case

Classes `ont:Material`, `ont:Process`, `ont:Quality`, `ont:SpatialRegion`, `ont:TemporalRegion`, and `ont:AbstractEntity` are defined with full disjointness axioms but have no properties, no subclasses relevant to the org chart, and no role in the stated purpose of showing who works for whom and who manages clients/vendors. This inflates the ontology with dead weight.

The range of `ont:reportsTo` is `ont:Agent`, which is far too broad. This allows a Person to report to a Vendor, a Community, or an ExternalCommunity, which is semantically incorrect for an org chart. The range should be restricted to `ont:Person` or at most `ont:Employee` / `ont:Manager`.

There is no explicit 'manages' or 'isManagedBy' supervisory property between Persons. The org chart use case requires modeling who manages whom (the managerial hierarchy), but the ontology only provides `ont:reportsTo` (the subordinate's perspective) without a named inverse property, making it harder to query 'who does Person X manage?'

The `ont:hasIndustry` datatype property has its domain set to `ont:Company`, but it would logically apply to any `ont:Organization` (e.g., a Vendor or ResearchInstitution also has an industry). This domain restriction is unnecessarily narrow and inconsistent with the broader class hierarchy.

No ontology-level metadata beyond a single `rdfs:comment` is provided. There is no `owl:versionInfo`, `dcterms:creator`, `dcterms:created`, `dcterms:title`, or `owl:imports` declaration, making the ontology difficult to manage, version, and discover.

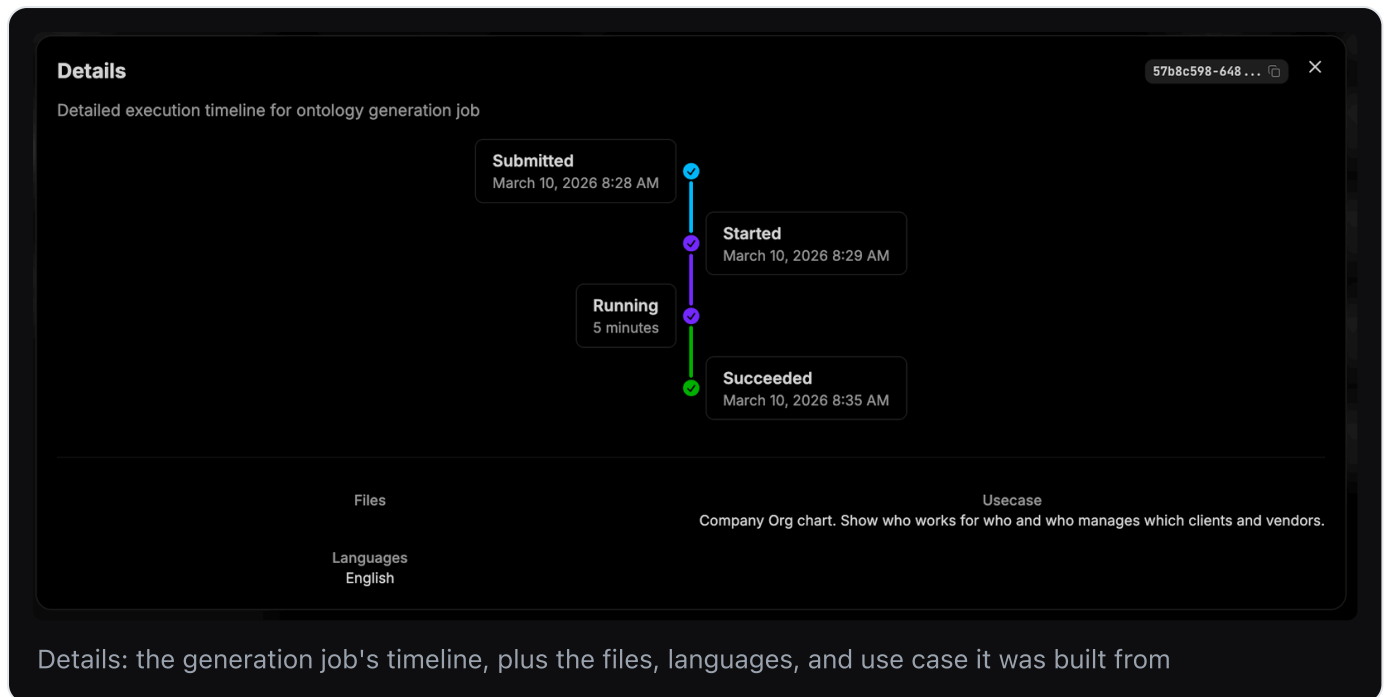
The ontology does not reuse any established vocabularies. The W3C Organization Ontology (ORG), FOAF, or schema.org already define Person, Organization, membership, and reporting relationships with broad community adoption. Reinventing these concepts creates interoperability barriers.

Evaluation → Weaknesses

The evaluation judges the ontology *against the use case you described*, which is why the use case is worth writing carefully — it is the yardstick. A common finding is over-engineering: the generator emits upper-ontology classes (`Material`, `Process`, `Quality`, `SpatialRegion`, `TemporalRegion`, `AbstractEntity`) that play no part in your actual use case and only add noise.

Details

The execution record of the generation job: a timeline of **Submitted** → **Started** → **Running** → **Succeeded** with timestamps and duration, the job ID, the **Files** it was built from, the **Languages** selected, and the **Usecase** text you typed in the wizard.



This is where to look when you have forgotten what a generated ontology was *for* — the use case is recorded here verbatim, and nowhere in the editor.

Three ways to get an ontology

1. **Build Ontology** — let Perseus generate one from documents you have already uploaded. Start here. Generating from a handful of representative documents gets you most of the way, and what remains is much easier to see with something concrete in front of you than from a blank ontology.
2. **Upload** — bring an existing `.ttl` file. Useful if you already maintain an ontology elsewhere, or you are importing a standard vocabulary.
3. **Open an existing one** and edit it. Perseus-generated ontologies are named `generated-ontology-<timestamp>.ttl` until you rename them.

Building an ontology from your files

Build Ontology opens a three-step wizard:

1. **Select files** — tick the workspace files to derive the ontology from. Perseus reads them to work out what classes and relationships your domain contains, so pick files that are representative rather than exhaustive.

Build ontology

Select files

Describe usecase

Select languages

Select files

Select files from your workspace to build an ontology from.

Q Search files...

☐ Org_part2.md1 month ago

☐ Org_part1.md1 month ago

☐ Events.md1 month ago

☐ People.md1 month ago

☐ Org.md1 month ago

☐ Hackersquad.md1 month ago

☐ 2024-06-11 HackNight.md1 month ago

☐ Adam Chan.md1 month ago

→ Configure usecase

Step 1 — Select files

2. Describe usecase — a free-text box describing what you are modelling and what you will use the graph for. This steers what the generator considers important enough to become a class, so be specific about the questions you want the finished graph to answer.

Build ontology

Select files

Describe usecase

Select languages

Describe usecase

Provide a description of the use case for the ontology.
See the [documentation](#) on ontology building and filling the use case field.

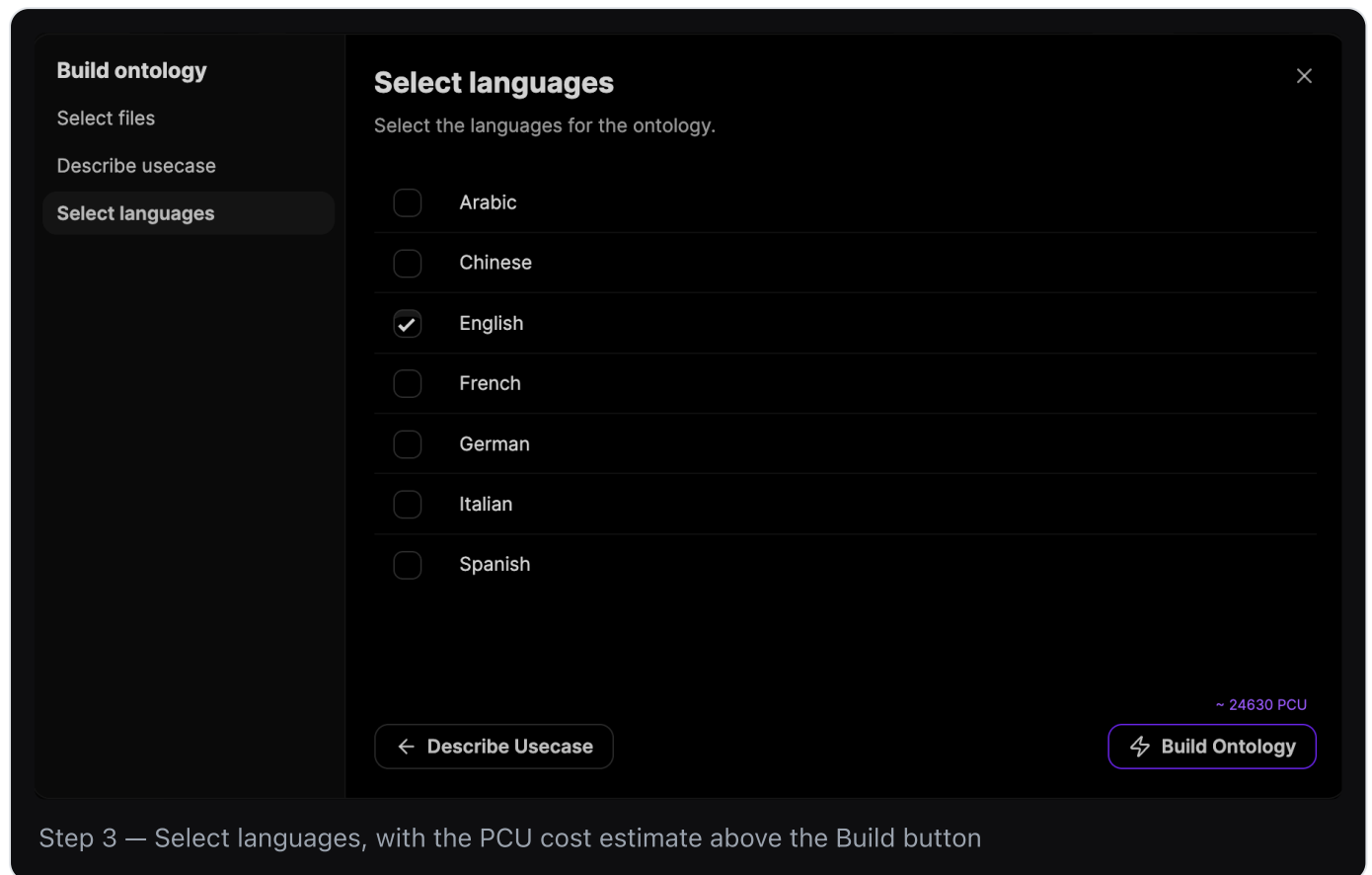
Describe your usecase

← Select Files

→ Select Languages

Step 2 — Describe usecase

3. Select languages — tick the languages the ontology should carry labels in (Arabic, Chinese, English, French, German, Italian, Spanish). This drives the language tags on the generated annotations.



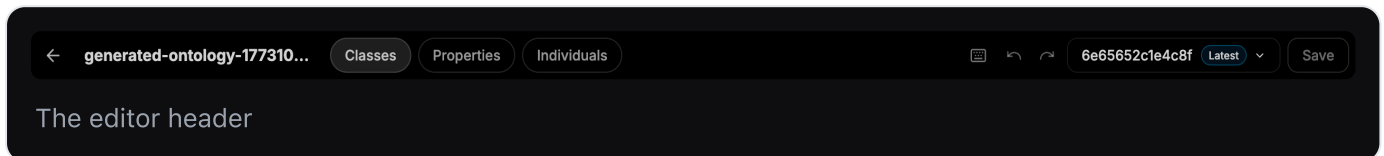
The screenshot shows a dark-themed interface for building an ontology. On the left is a sidebar with the title 'Build ontology' and three menu items: 'Select files', 'Describe usecase', and 'Select languages' (which is highlighted). The main area is titled 'Select languages' with a close button (X) in the top right. Below the title is the instruction 'Select the languages for the ontology.' A list of languages with checkboxes follows: Arabic, Chinese, English (checked), French, German, Italian, and Spanish. At the bottom left is a button with a left arrow and the text 'Describe Usecase'. At the bottom right is a button with a lightning bolt icon and the text 'Build Ontology'. Above the 'Build Ontology' button, the text '~ 24630 PCU' is displayed. Below the main interface area, a caption reads: 'Step 3 — Select languages, with the PCU cost estimate above the Build button'.

The final screen shows an estimated cost in **PCU** (Perseus Compute Units) before you commit. Click **Build Ontology** to run it. When it finishes, the new ontology appears in the library.

Expect to edit. A generated ontology is a strong first draft, not a finished artifact. Before you open the editor, open the card's **...** menu → **Evaluation** → **Recommendations**: the build has already reviewed its own output and told you what to fix. Working that list is the fastest way to a usable ontology.

4. The editor layout

Open an ontology and you get a full-screen editor.



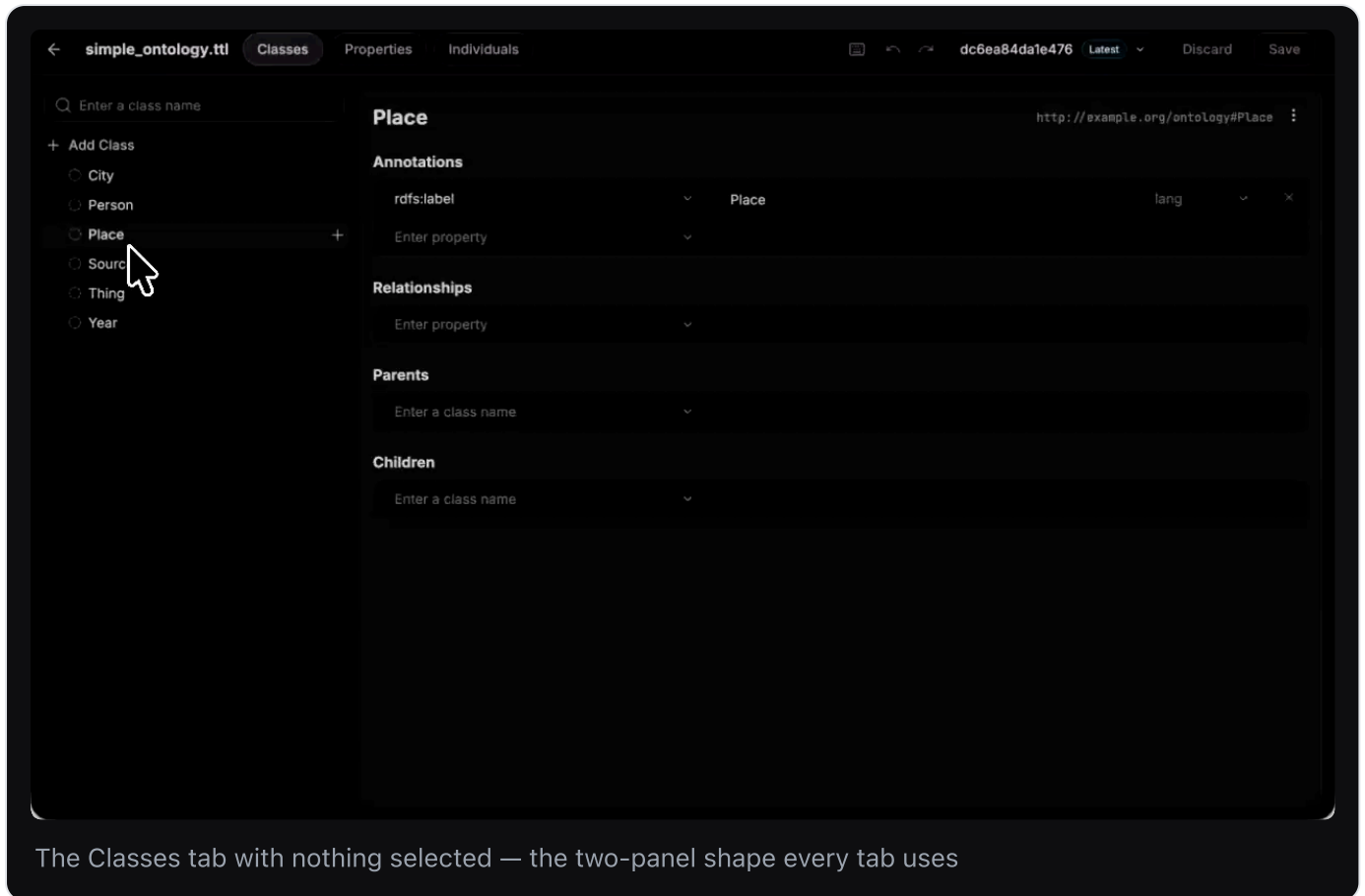
The editor header

The header, left to right:

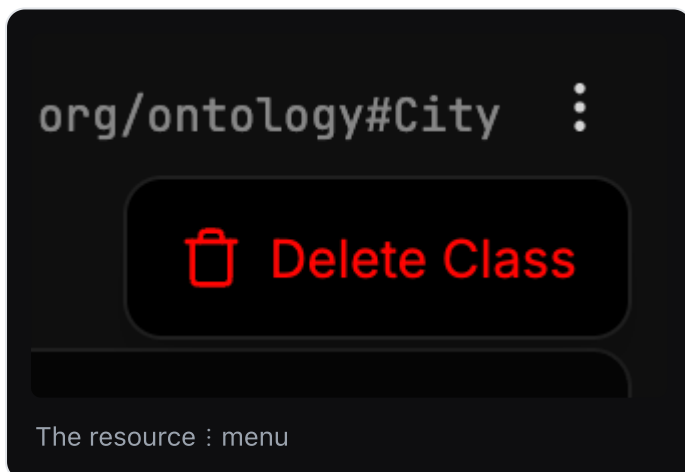
Element	What it does
←	Back to the Ontologies library.
Filename	The name of the ontology you have open.
Classes / Properties / Individuals	The three tabs. Each is a different view of the same ontology.
☰ icon	Opens the keyboard shortcuts modal.
↶ / ↷	Undo / redo.
Version chip	The current commit hash (e.g. <code>cd494f51312665</code>) with a Latest badge. Click the chevron for the version history dropdown.
Discard	Throw away all unsaved changes.
Save	Commit your changes with a message.

Below the header, every tab uses the same two-panel shape:

- **Left panel** — a searchable, filterable list of resources of that kind, plus an **Add** button at the top.
- **Right panel** — the detail form for whatever you selected. Empty until you select something ("No class selected — Select a class in the left panel to inspect it").



Every detail panel shows the resource's **name as an editable heading**, its **URI** in grey at the top-right, and a **menu** next to the URI.



That menu holds one action — **Delete Class** (or Delete Property / Delete Individual, depending on the tab). It is the only way to remove a resource other than the `⌘+⬇+⌫` shortcut.

Deleting is not a soft operation: if other resources referenced the deleted one — a class used as a property's domain, an individual on the far side of a relationship — those references go with it. Deletion only lands when you **Save**, so `⌘+Z` will walk it back and **Discard** will undo it wholesale.

Changes are held in memory as you work. Nothing is persisted until you **Save**.

5. Working with Classes

The **Classes** tab is where you define the types of thing in your domain.

Adding a class

Two ways:

- **+ Add Class** at the top of the left panel — creates a new top-level class.
- Hover any existing class and click the **+** that appears on its right — creates a new class as a *child of that one*. This is the fast way to build a hierarchy.

Either way you get a class named **Untitled**, selected, with a placeholder timestamp URI. Type over the heading in the right panel to name it. Perseus writes the name into `rdfs:label` and rewrites the URI to match.

Naming convention: singular, PascalCase — `Country` , not `countries` .

The class detail panel

Selecting a class shows four sections.

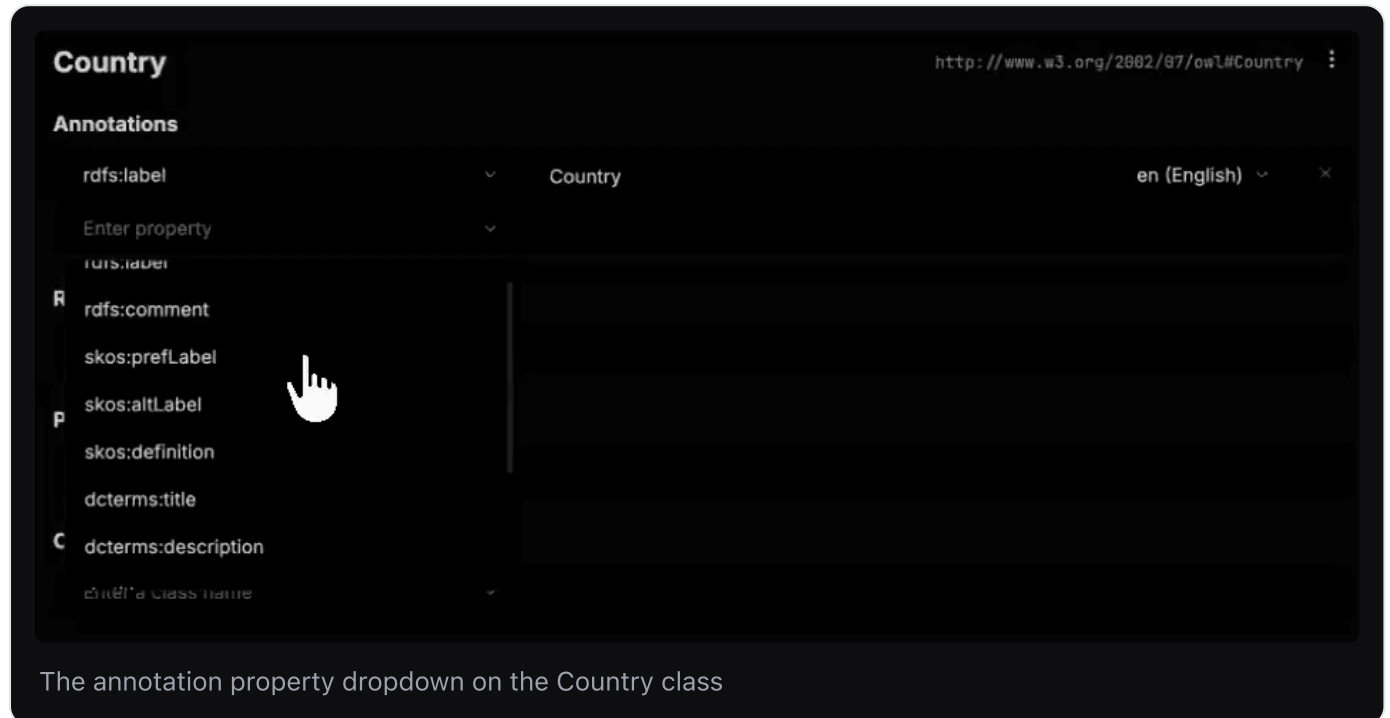
The screenshot shows the 'Agent' class detail panel in an ontology editor. The interface has a dark theme. At the top, there's a breadcrumb 'generated-ontology-177310...' and tabs for 'Classes', 'Properties', and 'Individuals'. The 'Classes' tab is active. On the right, there's a URI 'http://example.org/ontology#Agent' and a 'Save' button. The left panel shows a search bar 'Enter a class name' and a list of classes: 'Add Class', 'Abstract Entity', 'Agent' (selected), 'Material', 'Process', 'Quality', 'Spatial Region', 'Temporal Region', and 'test'. The main panel for 'Agent' has four sections: 'Annotations' with two entries (rdfs:label 'Agent' and rdfs:comment 'Entities capable of intentional action and decision-making, including persons and organizations.'), 'Relationships' with an 'Enter property' field, 'Parents' with an 'Enter a class name' field, and 'Children' with two entries ('Organization' and 'Person') and an 'Enter a class name' field.

The class detail panel. Agent has two annotations, no relationships, no parents, and two children

Annotations

Free-form documentation. Each row is: **property** + **value** + **language tag**.

Click the **Enter property** dropdown to pick the annotation type. The list runs `rdfs:label`, `rdfs:comment`, `skos:prefLabel`, `skos:altLabel`, `skos:definition`, `dcterms:title`, `dcterms:description`, `dc:title`, `dc:description`, `rdfs:seeAlso`, and `rdfs:isDefinedBy`. Type the value on the right, set the language on the far right, and add another row as needed. The `×` removes a row.



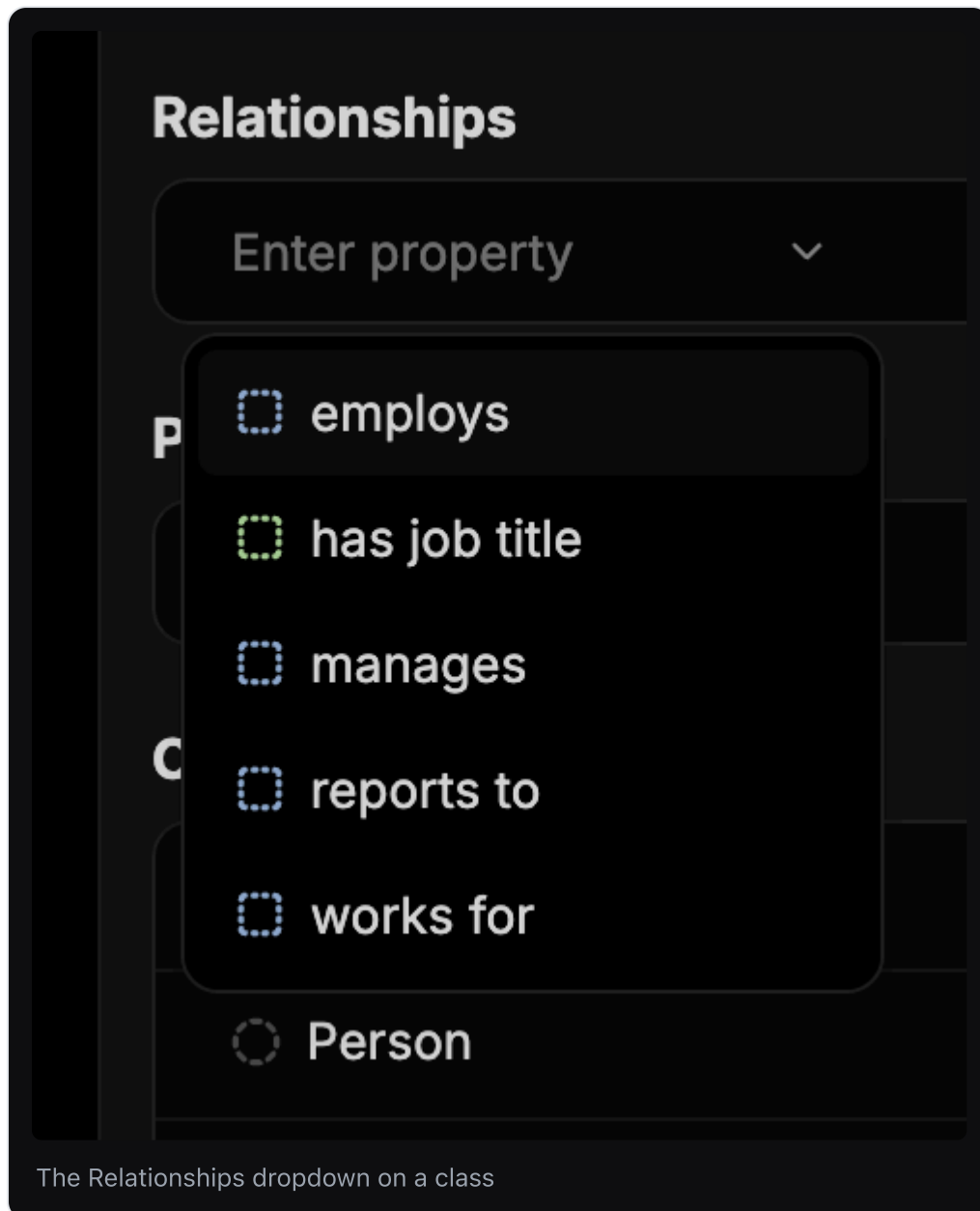
The annotation property dropdown on the Country class

A newly-named class arrives with `rdfs:label` already filled in. Add a description too — it is part of the prompt the extractor reads, so it directly affects extraction quality. See Annotation.

`skos:prefLabel` and `skos:altLabel` are worth knowing: the first is the preferred display name, the second holds synonyms. Putting known aliases on a class (`Firm`, `Corporation` as altLabels of `Company`) gives the extractor more surface to match against.

Relationships

The object properties that can start at this class. Opening the dropdown shows every property defined in the ontology (`ASSOCIATED_WITH`, `description`, `LOCATED_IN`, `MENTION`, `name`, `url` in the sample ontology), and it is a multi-select.

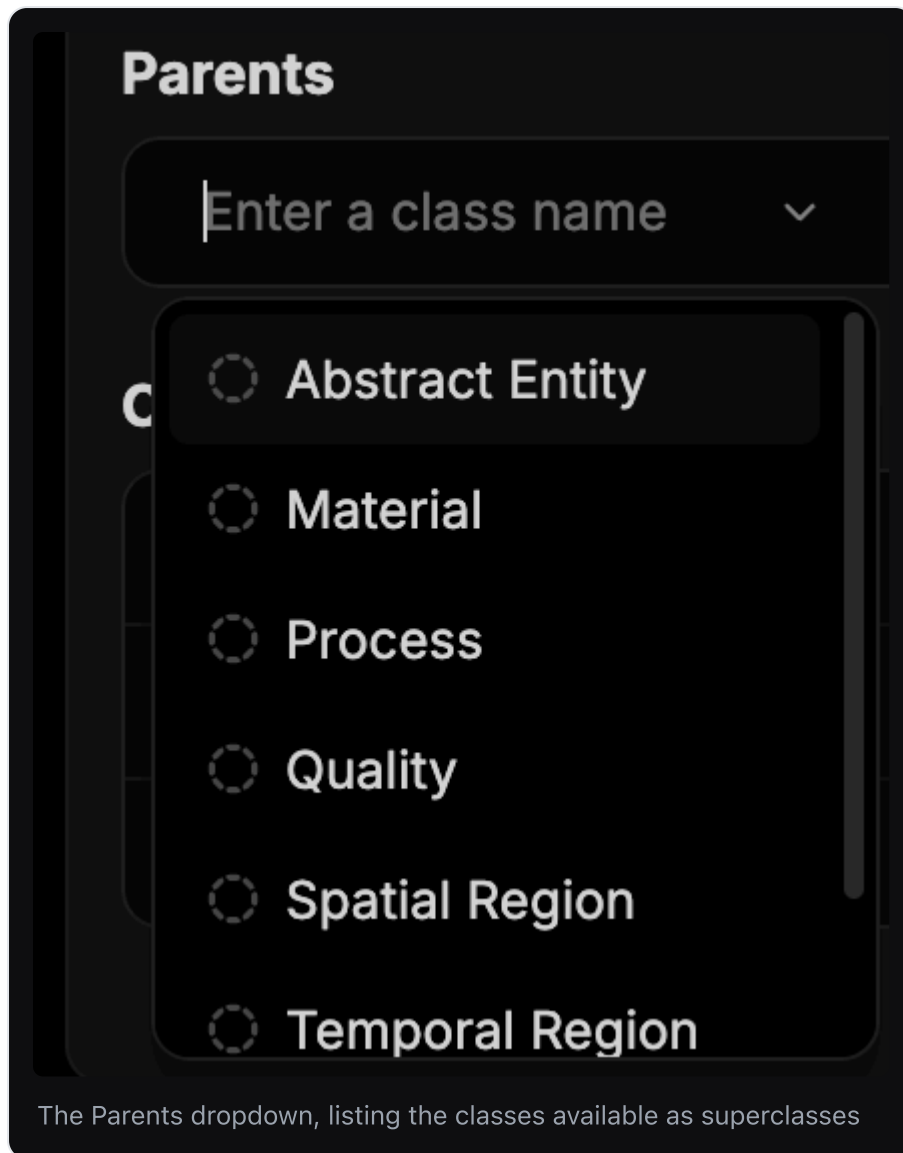


Adding a property here is the same as adding this class to that property's **domain** — you are saying "instances of this class may be the subject of this relationship." You can do it from either side; the Relationships box on the class is the convenient side when you are thinking class-first.

Parents and Children

The subclass hierarchy.

- **Parents** — the classes this one is a subclass of. Type or pick a class name. A class may have several parents.
- **Children** — the classes that are subclasses of this one.



The two are views of the same fact. Setting `City` 's parent to `Place` is identical to adding `City` to `Place` 's children; the other panel updates accordingly. Use whichever direction you are thinking in.

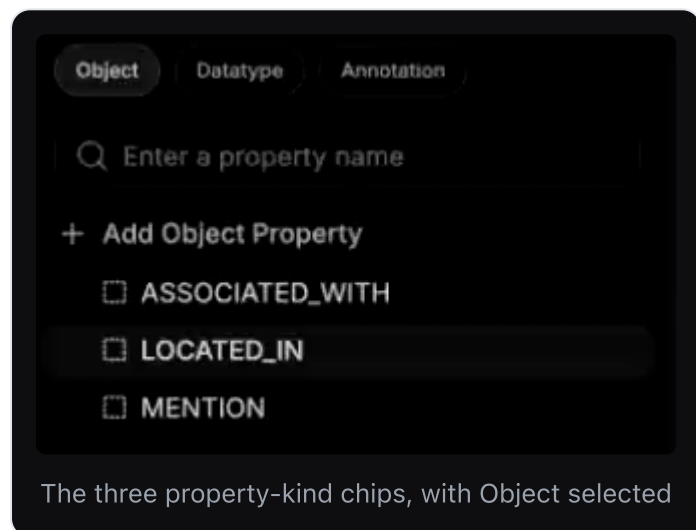
Parent/child means strict is-a, not "related to": everything true of the parent is true of the children. If you would not say "every X is a Y", do not make X a child of Y — model it as an object property between them instead. Treating is-a as a general-purpose "related to" is the most common modelling error.

6. Working with Properties

The **Properties** tab is where you define the edges.

The three property tabs

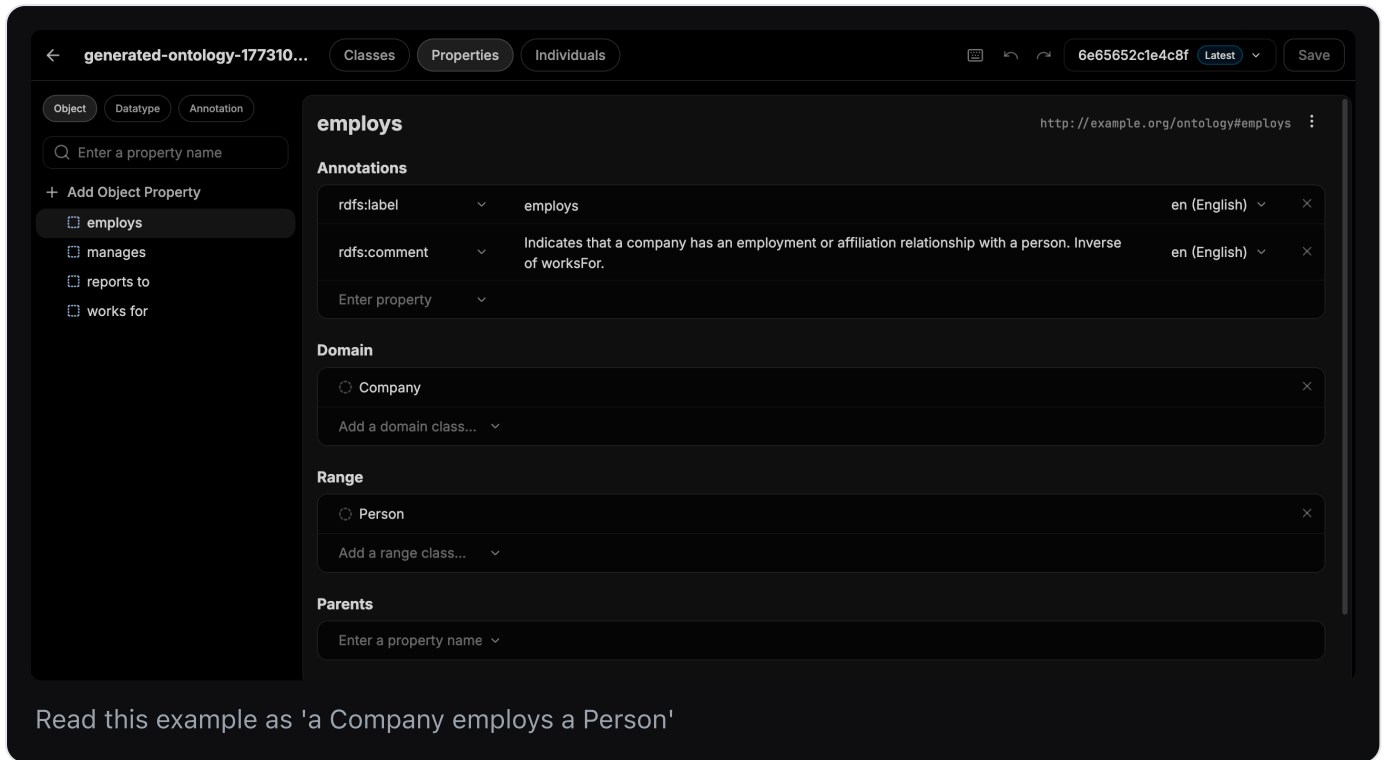
Three filter chips at the top of the left panel — **Object**, **Datatype**, **Annotation** — switch which kind of property you are looking at. The Add button relabels itself to match (+ Add Object Property , + Add Datatype Property , ...).



Object = entity-to-entity, datatype = entity-to-literal, annotation = documentation. See Property for choosing between them.

The property detail panel

Select a property and you get:



- **Annotations** — same widget as on classes. `rdfs:label` plus, ideally, an `rdfs:comment` explaining exactly when this relationship applies. In the sample data, `employs` carries the comment *"Indicates that a company has an employment or affiliation relationship with a person. Inverse of worksFor."* — that is the standard to aim for.
- **Domain** — the classes allowed on the *left* of this relationship. Add with **Add a domain class...**, remove with the `×`.
- **Range** — the classes allowed on the *right*. Add with **Add a range class...**.
- **Parents / Children** — the sub-property hierarchy. Rarely needed, but it lets you say e.g. that `manages` is a specialisation of `works with`; every `manages` triple then also counts as a `works with` triple.

Domain and range are the fields that do the real work — getting them right is most of what makes extraction precise. See Domain and Range.

LOCATED_IN

http://example.org/ontology#locatedIn

Annotations

rdfs:label

LOCATED_IN

lang

Enter property

Domain

City

Country

Add a domain class...

Range

City

Person

Add a range class...

Parents

Enter a property name

Children

Enter a property name

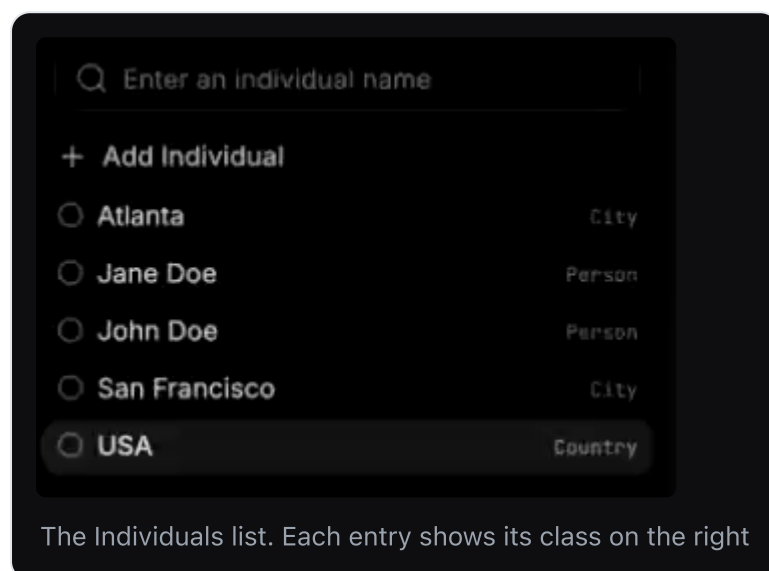
Adding a property

Click `+ Add Object Property` (or the datatype/annotation equivalent), name it in the heading, then set its domain and range. A property with an empty domain and range is legal but unconstrained, and will match far more loosely than you want. Start narrow and widen later.

Naming convention: the generated ontologies use two styles — `SCREAMING_SNAKE_CASE` for object properties (`LOCATED_IN` , `ASSOCIATED_WITH`) and `lowerCamelCase` or plain words for datatype properties (`name` , `url` , `has job title`). Neither is required. Pick one convention per ontology and hold to it.

7. Working with Individuals

The **Individuals** tab holds concrete, named entities. The left panel lists each individual with its class in grey on the right (`San Francisco` – `City`).



Adding an individual

Click `+ Add Individual` , then type its name over the heading. As with classes, it starts as an unnamed resource with a timestamp URI which settles once you name it.

An individual with no type is not useful, so the next step is always Types.

Types

The **Types** section is where you declare which class(es) this individual is an instance of. Pick from the class dropdown. Once set, the class appears next to the individual's name in the left panel.

This is the `rdf:type` triple — `:San_Francisco rdf:type :City` . An individual can have more than one type.

Relationships between individuals

The **Relationships** section is where you assert actual facts. Note how it differs from the identically-named section on a *class*:

- On a **class**, Relationships declares what is *possible* (schema).

- On an **individual**, Relationships states what is *true* (data).

Each row is two dropdowns: pick the **property** on the left, then **Pick an individual...** on the right.

The screenshot shows a web interface for an individual named "Jane Doe". At the top right is the URI: `http://www.w3.org/2002/07/owl#Jane_Doe`. The interface is divided into four main sections:

- Annotations:** A dropdown menu shows `rdfs:label` with the value "Jane Doe" and a language selector set to "en (English)". Below it is a text input "Enter property".
- Types:** A radio button is selected for "Person". Below it is a text input "Enter a class name".
- Relationships:** A dropdown menu shows `LOCATED_IN` with a radio button selected for "San Francisco". Below it is a text input "Enter property".
- Identity:** This section contains two sub-sections: "Same as" and "Different from", each with a dropdown menu labeled "Add an individual...".

At the bottom of the interface, a summary line reads: "The individual Jane Doe: typed as a Person, asserted LOCATED_IN San Francisco".

Jane Doe + LOCATED_IN + San Francisco creates exactly one triple:

```
:Jane_Doe :LOCATED_IN :San_Francisco .
```

The right-hand dropdown only lists individuals that already exist, so create the target individual before the relationship that points at it.

Identity: Same as / Different from

Two fields at the bottom:

- **Same as** (`owl:sameAs`) — asserts that this individual and another are *the same real-world thing under two names*. `:NYC owl:sameAs :New_York_City` . Anything true of one is now true of the other. This is the mechanism for reconciling duplicates that extraction produced from different documents.
- **Different from** (`owl:differentFrom`) — asserts they are definitively *not* the same thing, even though they look similar. Useful for pinning apart two people with the same name, and for preventing a downstream deduplication step from merging them.

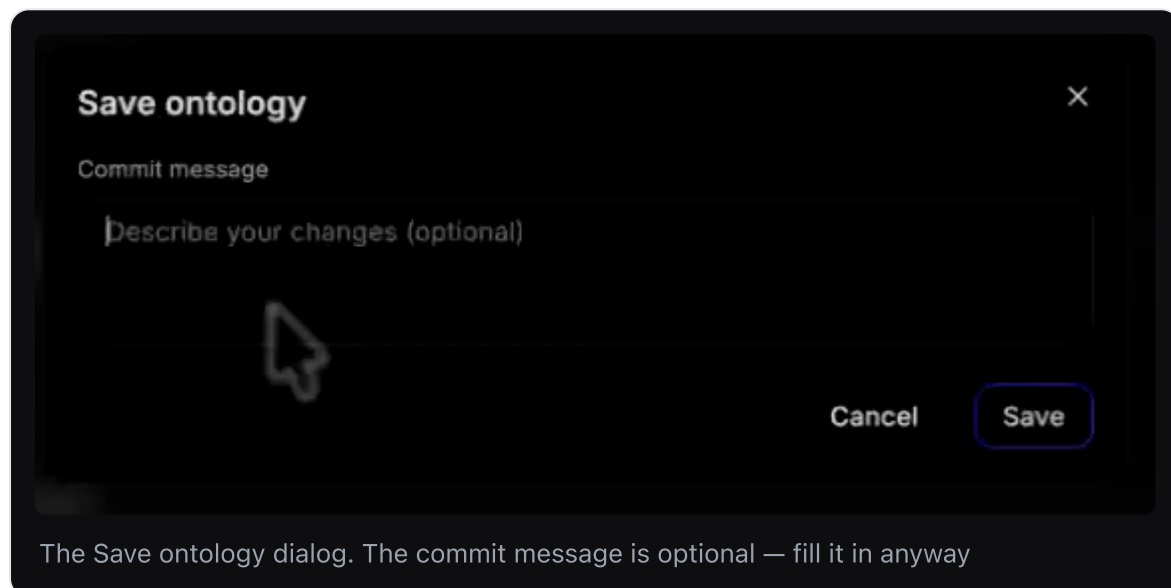
Left unset, neither is assumed. Two individuals with different URIs are not automatically considered distinct — this is the *open-world assumption* that RDF inherits, and it is why `differentFrom` needs to exist at all.

8. Saving and version history

Perseus versions ontologies like a git repository. Every save is a commit with a hash and a message, and you can browse back through them.

Save with a commit message

Click **Save** (or `⌘S`). A **Save ontology** dialog appears with a **Commit message** field. Type a short description of what changed — *"Added Country"*, *"People-[LOCATED_IN]→City"* — and click **Save**.



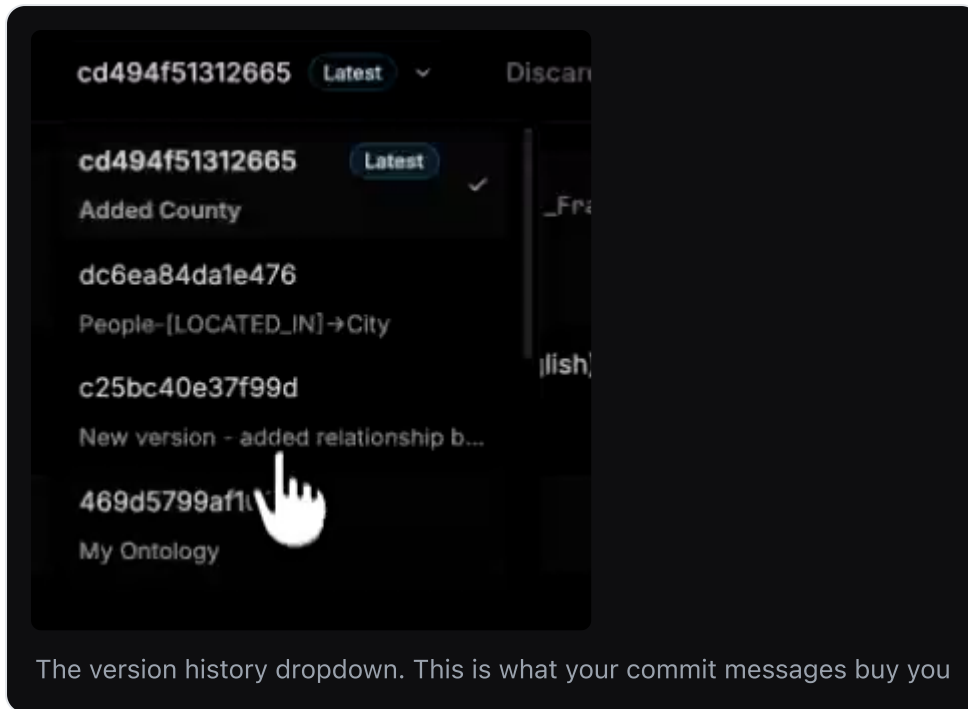
The version chip in the header updates to the new hash and keeps the **Latest** badge.

Save is a commit, not an autosave. Nothing persists until you click it, and navigating away or hitting **Discard** loses everything since the last one. Save early and often — the version history is cheap, and it is the only path back.

The message field is technically **optional**, and Perseus will commit without one. Fill it in anyway. These messages *are* the version history dropdown, and a list of unlabelled hashes is not a history you can navigate.

Browsing previous versions

Click the chevron next to the version chip. You get the full commit list, newest first — each entry showing its hash and its commit message, with **Latest** marked and a check against the version you are currently viewing. Select any entry to load that version of the ontology.



Discard

Discard throws away every unsaved change and returns you to the last saved version. It does not delete anything that has been committed. There is no undo for Discard — but there is `⌘Z` for individual mistakes, which is usually what you actually want.

9. Keyboard shortcuts

Open the shortcuts modal with the  icon in the header, or   .

Shortcuts

Shortcuts details

Open the keyboard shortcuts modal

  ?

Undo

Undo the last change

 Z

Redo

Redo the last undone change

  Z

Save

Save the current changes

 S

Focus search

Search

Focus the resource filter input

 G

Create new resource

Create a new resource in the ontology depending on the resource tab.

  N

Edit resource name

Focus the selected resource name input for editing.

F2

Delete resource

Delete the selected resource from ontology.

The Shortcuts modal

Action	Shortcut	Notes
Shortcuts details	⌘↑?	Opens this modal.
Undo	⌘Z	Undo the last change.
Redo	⌘↑Z	Redo the last undone change.
Save	⌘S	Opens the commit-message dialog.
Focus search	⌘G	Jumps to the left-panel filter input.
Create new resource	⌘↑N	Creates a class, property, or individual — whichever tab you are on.
Edit resource name	F2	Focuses the selected resource's name for renaming.
Delete resource	⌘↑⌫	Deletes the selected resource from the ontology.

On Windows and Linux, substitute `Ctrl` for `⌘`.

`⌘G` → type → `⌘↑N` → type name → `F2` is the fast loop for building out a hierarchy without touching the mouse.

10. A complete worked example

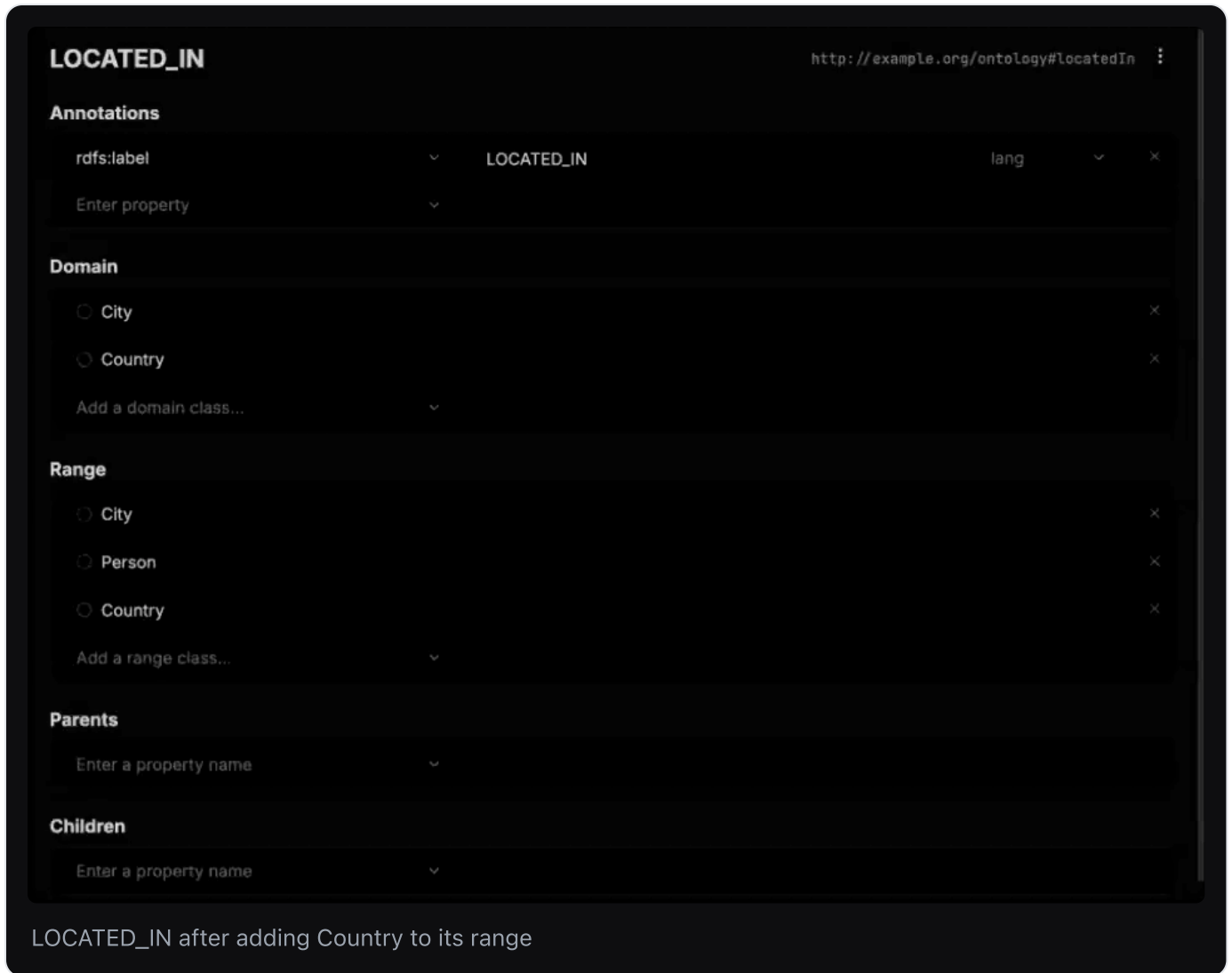
This exercise walks the whole editor — a class, a property, individuals, facts, a commit — in six steps. Open any ontology you have; a generated one from **Build Ontology** works well, since it already has classes and properties to build on.

The walkthrough below uses a small ontology whose classes include `City` and `Person` and whose object properties include `LOCATED_IN`. Substitute your own names as you go: the shape of each step is the same whatever your domain is. The goal is to add a class that is missing, wire it into an existing relationship, and then state two facts using it.

1. Add the missing class. Classes tab → `+ Add Class` → the new *Untitled* class appears. Type a name over the heading — here, `Country`. Perseus fills in `rdfs:label = "Country"` with language `en` (`English`) and settles the URI to `owl#Country`.

2. Document it. In **Annotations**, add a second row: `dcterms:description`, with a sentence explaining what counts as a Country in this domain.

3. Let the new class participate in a relationship. Properties tab → **Object** → select an existing property, here `LOCATED_IN`. Under **Range**, click **Add a range class...** and add `Country`. Its range becomes `{City, Person, Country}` and its domain is `{City, Country}` — so a `City` may now be `LOCATED_IN` a `Country`.



4. Create two individuals. Individuals tab → + Add Individual → name it `Jane Doe` → under **Types**, pick `Person`. Repeat: + Add Individual → `USA` → under **Types**, pick `Country`. The left panel now shows each individual with its class beside it.

5. Assert the facts. Assuming an individual `San Francisco` of class `City` already exists:

- Select `Jane Doe` → **Relationships** → property `LOCATED_IN`, individual `San Francisco`.
- Select `San Francisco` → **Relationships** → property `LOCATED_IN`, individual `USA`.

6. Commit. Save → commit message "*Added Country*" → **Save**. The version chip changes to a fresh hash carrying the **Latest** badge, and the new commit is at the top of the version dropdown.

In Turtle, what you just built is:

```
:Country      rdf:type      owl:Class ;  
              rdfs:label    "Country"@en .  
  
:LOCATED_IN   rdfs:range    :City, :Person, :Country .  
  
:Jane_Doe     rdf:type      :Person ;  
              :LOCATED_IN   :San_Francisco .  
  
:San_Francisco rdf:type      :City ;  
              :LOCATED_IN   :USA .  
  
:USA          rdf:type      :Country .
```